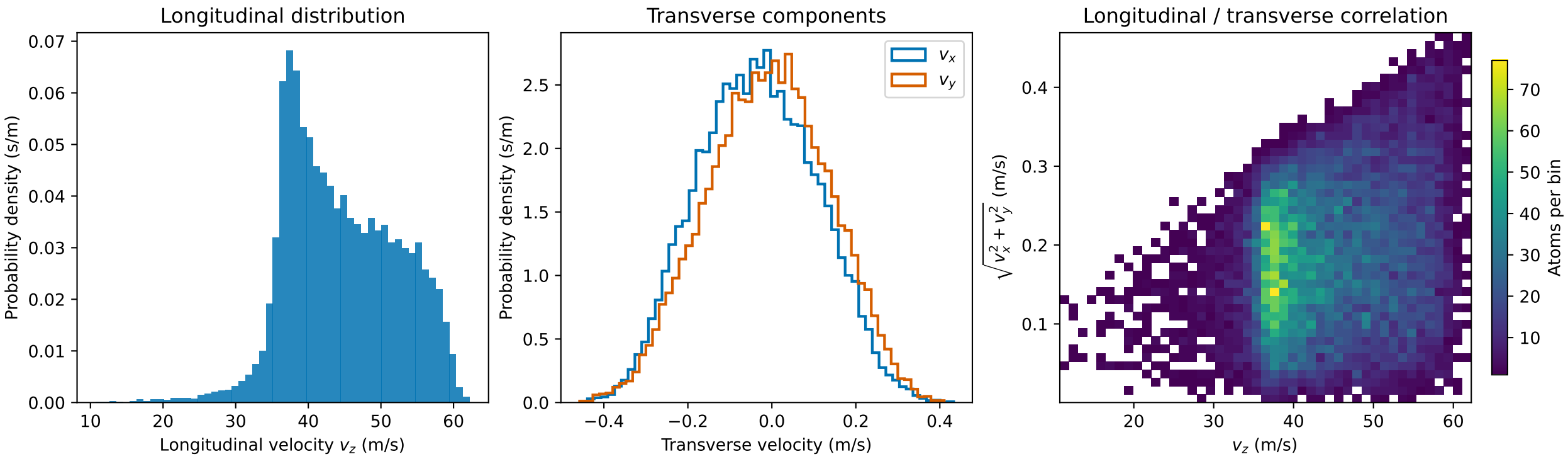

Velocity distribution after the optimized 2D MOT
Longitudinal distribution
Probability density (s/m)
Longitudinal velocity v_z (m/s)
Transverse components
Probability density (s/m)
v_x
v_y
Transverse velocity (m/s)
Longitudinal / transverse correlation
$\sqrt{v_x^2 + v_y^2}$ (m/s)
v_z (m/s)
Atoms per bin
16,883 surviving atoms
$s_0 = 1.3$, $\Delta/\Gamma = -0.919661$, magnet radius 46.7641 mm.